FRONTEND DEVELOPMENT

&

Bootstrap 5

Complete Student Study Notes

Topics Covered

1. Responsive Web Design
2. Bootstrap Introduction
3. Grid System
4. Bootstrap Components
5. Customizing Bootstrap

Learning Outcomes

✓ Build responsive layouts
✓ Use Bootstrap grid & components
✓ Create UI with Bootstrap
✓ Customize Bootstrap themes
✓ Apply RWD principles

01 Responsive Web Design Principles

Building websites that work on any screen size

1.1 What is Responsive Web Design?

Responsive Web Design (RWD) is an approach to web development that makes web pages render well on a variety of devices and window or screen sizes — from desktop monitors to mobile phones. The goal is to ensure users get an optimal viewing and interaction experience.

The three core techniques of RWD are:

- **Fluid Grids:** Layout elements are sized in relative units (%, em, rem) rather than fixed pixels
- **Flexible Images:** Images scale within their containing elements to avoid overflow
- **CSS Media Queries:** Different CSS rules applied based on viewport size/device characteristics

> *RWD was introduced by Ethan Marcotte in 2010. Today, over 60% of web traffic comes from mobile devices — making RWD an essential skill for every frontend developer.*

1.2 Viewport and the Meta Tag

The viewport meta tag tells the browser how to control the page's dimensions and scaling. Without it, mobile browsers render pages at desktop width and then scale down.

Essential viewport meta tag (place in <head>):

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<!-- width=device-width : sets width to screen width -->
<!-- initial-scale=1.0 : sets initial zoom level to 1 -->
```

- **width=device-width:** Matches the screen width of the device
- **initial-scale=1.0:** Sets default zoom level when page first loads
- **user-scalable=no:** Optional — prevents users from zooming (not recommended for accessibility)

1.3 CSS Media Queries

Media queries are the backbone of responsive design. They allow you to apply different CSS rules depending on the characteristics of the device (screen width, orientation, resolution, etc.).

Syntax:

```
@media (condition) {
  /* CSS rules applied when condition is true */
}

/* Example: Change layout for screens <= 768px (tablets/phones) */
@media (max-width: 768px) {
  .container { flex-direction: column; }
  .sidebar { display: none; }
```

```
  h1 { font-size: 1.5rem; }
}

/* Min-width approach (Mobile First) */
@media (min-width: 992px) {
  .container { display: flex; }
}
```

Common Breakpoints

- Mobile (phones): max-width: 576px
- Tablets: 577px – 768px
- Small desktops: 769px – 992px
- Large desktops: 993px – 1200px
- Extra-large: 1201px and above

1.4 Mobile-First vs Desktop-First Design

Mobile-First (Recommended)	Desktop-First
Design for mobile screen first Uses min-width media queries Better performance on mobile Default approach in Bootstrap 5	Design for large screens first Uses max-width media queries Easier for complex layouts Older/traditional approach

Always use Mobile-First approach when working with Bootstrap 5. It results in leaner, faster code and aligns with Bootstrap's built-in breakpoint system.

1.5 Flexbox & CSS Grid Basics

Modern responsive layouts rely heavily on CSS Flexbox and CSS Grid — both are natively supported in Bootstrap 5 through utility classes.

```
/* Flexbox */
.container { display: flex; flex-wrap: wrap; gap: 16px; }
.item { flex: 1 1 200px; } /* grow, shrink, basis */

/* CSS Grid */
.grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(250px, 1fr)); gap:
24px; }
```

02 Introduction to Bootstrap 5

Framework overview, features, and advantages

2.1 What is Bootstrap?

Bootstrap is the world's most popular open-source CSS framework for building responsive, mobile-first websites and web applications. Originally created by Twitter engineers Mark Otto and Jacob Thornton in 2011, Bootstrap provides a comprehensive collection of pre-built CSS components, JavaScript plugins, and a powerful grid system.

> *Bootstrap 5 was released in May 2021. It dropped jQuery dependency (now uses vanilla JavaScript), improved the grid system, added new utilities, and introduced the utility API.*

Key Facts about Bootstrap 5

- Current Version: Bootstrap 5.3+
- License: MIT (free and open source)
- Dependencies: None (no jQuery required)
- CDN Available: Yes — usable without installation
- Browser Support: Modern browsers + IE not supported

2.2 Features of Bootstrap 5

#	Feature	Description
1	**Responsive Grid**	12-column fluid grid that adapts to any screen size automatically
2	**Pre-built Components**	Ready-to-use UI elements: navbars, buttons, cards, forms, modals, and more
3	**JavaScript Plugins**	Interactive components: dropdowns, carousels, tooltips, modals, etc.
4	**Utility Classes**	Hundreds of helper classes for spacing, typography, colors, display, and more
5	**Sass Variables**	Fully customizable via Sass — change colors, fonts, sizes project-wide
6	**RTL Support**	Built-in Right-to-Left text direction support (new in Bootstrap 5)
7	**CSS Custom Properties**	Native CSS variables for real-time theme customization
8	**No jQuery**	Pure vanilla JavaScript — lighter, faster, and more modern

2.3 Setting Up Bootstrap 5

Method 1: CDN (Quickest for Learning)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Bootstrap Page</title>
  <!-- Bootstrap 5 CSS (place in <head>) -->
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
        rel="stylesheet">
</head>
<body>
  <!-- Your content here -->

  <!-- Bootstrap 5 JS Bundle (place before </body>) -->
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js">
  </script>
</body>
</html>
```

Method 2: npm Install

```
# Install via npm
npm install bootstrap

# Import in your JavaScript/SCSS
import 'bootstrap/dist/css/bootstrap.min.css';
import 'bootstrap';
```

> *For learning and prototyping, use the CDN method. For production projects, install via npm so you can customize with Sass and tree-shake unused components.*

2.4 Advantages of Bootstrap

- **Speed:** Drastically reduces development time with ready-made components
- **Consistency:** Uniform look and feel across all browsers and devices
- **Community:** Massive community, thousands of themes, plugins, and tutorials
- **Responsive by Default:** Mobile-first grid system built in from the start
- **Documentation:** Excellent official docs at getbootstrap.com
- **Customizable:** Sass-based customization for project-specific designs

03 Bootstrap Grid System

Responsive 12-column layout for any design

3.1 Understanding the Grid System

Bootstrap's grid system uses a series of containers, rows, and columns to layout and align content. It is built with Flexbox and is fully responsive. The grid divides the available width into 12 equal columns.

The three required elements:

- **.container (or .container-fluid):** Wraps the grid and provides horizontal padding
- **.row:** Creates horizontal groups of columns using negative margins
- **.col-*:** Individual columns — must be inside a .row

Basic Grid Structure

```
<div class="container">
  <div class="row">
    <div class="col-md-4">Column 1</div>
    <div class="col-md-4">Column 2</div>
    <div class="col-md-4">Column 3</div>
  </div>
</div>
<!-- Three equal columns on medium screens and above -->
<!-- Each col-md-4 = 4/12 = 1/3 of the row width -->
```

3.2 Grid Breakpoints

Breakpoint	Prefix	Min Width	Usage
Extra Small	col-	< 576px	Phones
Small	col-sm-	≥ 576px	Large phones
Medium	col-md-	≥ 768px	Tablets
Large	col-lg-	≥ 992px	Desktops
X-Large	col-xl-	≥ 1200px	Wide screens
XX-Large	col-xxl-	≥ 1400px	Ultra wide

Columns stack vertically (100% width) on screens BELOW their breakpoint. For example, col-md-6 takes 50% on medium and larger screens, but 100% width on small/extra-small screens.

3.3 Column Sizing Examples

Equal Width Columns

```
<!-- Auto-equal columns (Bootstrap calculates width) -->
<div class="row">
  <div class="col">1 of 3</div>  <!-- ~33.3% each -->
  <div class="col">2 of 3</div>
  <div class="col">3 of 3</div>
</div>
```

Different Width Columns

```
<!-- 1/3 + 2/3 layout -->
<div class="row">
  <div class="col-md-4">Sidebar (33%)</div>
  <div class="col-md-8">Main Content (66%)</div>
</div>
```

Responsive Columns (Different on Each Screen)

```
<!-- Full on mobile, half on tablets, 1/3 on desktops -->
<div class="row">
  <div class="col-12 col-sm-6 col-md-4">
    Card 1
  </div>
  <div class="col-12 col-sm-6 col-md-4">
    Card 2
  </div>
  <div class="col-12 col-sm-12 col-md-4">
    Card 3
  </div>
</div>
```

3.4 Gutters and Spacing

Gutters are the gaps between columns. Bootstrap 5 provides gutter utility classes (g-*, gx-*, gy-*) to control horizontal and vertical gaps.

```
<!-- g-* controls both horizontal and vertical gutters -->
<div class="row g-3">  <!-- gutter size 3 = 1rem -->
  <div class="col-md-6"><div class="p-3 bg-light">Box</div></div>
  <div class="col-md-6"><div class="p-3 bg-light">Box</div></div>
</div>

<!-- gx-* for horizontal only, gy-* for vertical only -->
<div class="row gx-5 gy-2">
  ...
</div>

<!-- Remove gutters completely -->
<div class="row g-0"> ... </div>
```

3.5 Offset, Order & Nesting

Column Offset

```
<!-- Push column to the right with offset -->
<div class="row">
  <div class="col-md-4 offset-md-4">Centered column</div>
</div>
```

Changing Column Order

```
<div class="row">
  <div class="col order-2">Shown second</div>
  <div class="col order-1">Shown first</div>
</div>
```

Nested Grids

```
<div class="col-md-9">
  <div class="row">  <!-- Nested row inside a column -->
    <div class="col-md-6">Nested col 1</div>
    <div class="col-md-6">Nested col 2</div>
  </div>
</div>
```

> *The columns in a nested row also span 12 units — relative to their parent column, not the whole page. You can nest grids as deeply as needed.*

04 Bootstrap Components

Buttons, Forms, Navbars, Cards & more

4.1 Component Overview

Component	Class/Tag	Purpose
Primary Button	btn btn-primary	Main call-to-action
Outline Button	btn btn-outline-*	Secondary action
Navbar	navbar navbar-expand-lg	Responsive nav
Card	card / card-body	Content container
Form Control	form-control	Input styling
Badge	badge bg-*	Labels & counts
Alert	alert alert-*	User messages

4.2 Buttons

Bootstrap buttons are created using the .btn class plus a variant class. They can be applied to <button>, <a>, or <input> elements.

```
<!-- Solid buttons -->
<button class="btn btn-primary">Primary</button>
<button class="btn btn-secondary">Secondary</button>
<button class="btn btn-success">Success</button>
<button class="btn btn-danger">Danger</button>
<button class="btn btn-warning">Warning</button>
<button class="btn btn-info">Info</button>
<button class="btn btn-dark">Dark</button>

<!-- Outline buttons (transparent background) -->
<button class="btn btn-outline-primary">Outline</button>

<!-- Button sizes -->
<button class="btn btn-primary btn-lg">Large</button>
<button class="btn btn-primary btn-sm">Small</button>

<!-- Full width button -->
<button class="btn btn-primary w-100">Full Width</button>

<!-- Disabled state -->
<button class="btn btn-primary" disabled>Disabled</button>
```

4.3 Navigation Bar (Navbar)

The navbar component provides a responsive navigation header. It collapses into a hamburger menu on smaller screens.

```
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
  <div class="container">
    <!-- Brand/Logo -->
    <a class="navbar-brand" href="#">MySite</a>

    <!-- Hamburger toggle (for mobile) -->
    <button class="navbar-toggler" type="button"
      data-bs-toggle="collapse" data-bs-target="#navMenu">
      <span class="navbar-toggler-icon"></span>
    </button>

    <!-- Collapsible nav links -->
    <div class="collapse navbar-collapse" id="navMenu">
      <ul class="navbar-nav ms-auto">
        <li class="nav-item"><a class="nav-link active" href="#">Home</a></li>
        <li class="nav-item"><a class="nav-link" href="#">About</a></li>
        <li class="nav-item"><a class="nav-link" href="#">Contact</a></li>
      </ul>
    </div>
  </div>
</nav>
```

- **navbar-dark bg-dark:** Dark background with white text (use navbar-light bg-light for inverse)
- **navbar-expand-lg:** Collapse below lg breakpoint, expand on lg+
- **ms-auto:** Pushes nav links to the right using margin-start: auto

4.4 Cards

Cards are flexible content containers with optional header, footer, images, and body sections. They're one of Bootstrap's most versatile components.

```
<div class="card" style="width: 18rem;">
  <!-- Card Image (optional) -->
  <img src="image.jpg" class="card-img-top" alt="...">

  <div class="card-body">
    <h5 class="card-title">Card Title</h5>
    <p class="card-text">Some quick example text.</p>
    <a href="#" class="btn btn-primary">Go somewhere</a>
  </div>

  <!-- Optional footer -->
  <div class="card-footer text-muted">
    2 days ago
  </div>
</div>

<!-- Card Grid: Multiple cards in a row -->
<div class="row row-cols-1 row-cols-md-3 g-4">
  <div class="col"><div class="card h-100">...</div></div>
  <div class="col"><div class="card h-100">...</div></div>
  <div class="col"><div class="card h-100">...</div></div>
</div>
```

4.5 Forms

Bootstrap provides styled form controls, labels, validation states, and layout helpers for building clean, accessible forms.

```
<form>
  <!-- Text Input -->
  <div class="mb-3">
    <label for="email" class="form-label">Email address</label>
    <input type="email" class="form-control" id="email"
           placeholder="name@example.com">
    <div class="form-text">We'll never share your email.</div>
  </div>

  <!-- Password Input -->
  <div class="mb-3">
    <label for="password" class="form-label">Password</label>
    <input type="password" class="form-control" id="password">
  </div>

  <!-- Select Dropdown -->
  <div class="mb-3">
    <label for="country" class="form-label">Country</label>
    <select class="form-select" id="country">
      <option selected>Choose...</option>
      <option>India</option>
      <option>USA</option>
    </select>
  </div>

  <!-- Checkbox -->
  <div class="form-check mb-3">
    <input class="form-check-input" type="checkbox" id="terms">
    <label class="form-check-label" for="terms">
      I agree to the Terms
    </label>
  </div>

  <button type="submit" class="btn btn-primary">Submit</button>
</form>
```

4.6 Other Key Components

Alerts

```
<div class="alert alert-success" role="alert">
  Operation successful! Click to <a href="#" class="alert-link">view details</a>.
</div>
<div class="alert alert-danger alert-dismissible fade show">
  Error occurred!
  <button type="button" class="btn-close" data-bs-dismiss="alert"></button>
</div>
```

Badges

```
<h2>Notifications <span class="badge bg-primary">4</span></h2>
<span class="badge bg-success">Active</span>
<span class="badge bg-danger rounded-pill">99+</span>
```

Modal Dialog

```
<!-- Trigger button -->
<button class="btn btn-primary" data-bs-toggle="modal" data-bs-target="#myModal">
  Open Modal
</button>
```

```
<!-- Modal HTML -->
<div class="modal fade" id="myModal" tabindex="-1">
  <div class="modal-dialog">
    <div class="modal-content">
      <div class="modal-header">
        <h5 class="modal-title">Modal Title</h5>
        <button type="button" class="btn-close" data-bs-dismiss="modal"></button>
      </div>
      <div class="modal-body">Modal content goes here.</div>
      <div class="modal-footer">
        <button class="btn btn-secondary" data-bs-dismiss="modal">Close</button>
        <button class="btn btn-primary">Save changes</button>
      </div>
    </div>
  </div>
</div>
```

05 Customizing Bootstrap

Themes, colors, and Sass customization

5.1 Bootstrap Utility Classes

Before diving into Sass customization, it's important to know that Bootstrap provides hundreds of utility classes for quick styling without writing custom CSS.

Spacing Utilities

```
<!-- m = margin, p = padding -->
<!-- t=top, b=bottom, s=start/left, e=end/right, x=horizontal, y=vertical -->
<!-- 0-5 for size (0=0rem, 1=0.25rem, 2=0.5rem, 3=1rem, 4=1.5rem, 5=3rem) -->

<div class="mt-3 mb-4 px-5 py-2">   <!-- margin-top:1rem, margin-bottom:1.5rem, etc. -->
<div class="ms-auto">                <!-- margin-start: auto (push right) -->
<div class="m-0 p-0">               <!-- remove all margin and padding -->
```

Typography Utilities

```
<p class="text-primary fw-bold fs-4">Bold Blue Text size 4</p>
<p class="text-center text-uppercase text-muted">Centered Uppercase Gray</p>
<p class="text-decoration-underline fst-italic">Underlined Italic</p>

<!-- Font sizes: fs-1 (largest) to fs-6 (smallest) -->
<!-- Font weight: fw-bold, fw-bolder, fw-normal, fw-light -->
```

Display & Flexbox Utilities

```
<div class="d-flex justify-content-between align-items-center">
  <span>Left item</span>
  <span>Right item</span>
</div>

<!-- d-none hides element, d-md-block shows from md breakpoint -->
<div class="d-none d-md-block">Visible only on md+</div>
```

5.2 CSS Custom Properties (Variables)

Bootstrap 5 exposes CSS custom properties (variables) that you can override globally without compiling Sass. This is the easiest customization method.

```
/* Override Bootstrap CSS variables in your stylesheet */
:root {
  --bs-primary: #8B5CF6;     /* Change primary color to purple */
  --bs-secondary: #10B981;   /* Change secondary to green */
  --bs-font-sans-serif: 'Poppins', sans-serif;  /* Custom font */
  --bs-border-radius: 0.75rem;  /* Rounder corners */
  --bs-body-bg: #F8FAFC;     /* Light gray body background */
  --bs-body-color: #1E293B;  /* Dark text color */
}
```

> *CSS variable overrides work at runtime and don't require any build tools. Just add your overrides in a custom CSS file AFTER the Bootstrap CDN link.*

5.3 Customizing with Sass

For deeper customization, Bootstrap uses Sass (SCSS). You can override Bootstrap's Sass variables before importing Bootstrap to change colors, spacing, typography, breakpoints, and more globally.

Setting Up Sass Customization

```
// custom.scss — your customization file

// STEP 1: Override Bootstrap Sass variables
$primary:          #8B5CF6;   // Purple instead of blue
$secondary:        #10B981;   // Green
$body-bg:          #F8FAFC;
$body-color:       #1E293B;
$border-radius:    0.75rem;
$font-size-base:   1rem;
$font-family-base: 'Poppins', sans-serif;

// Customize spacing scale
$spacer:           1.25rem;

// STEP 2: Import Bootstrap AFTER your overrides
@import '../node_modules/bootstrap/scss/bootstrap';

// STEP 3: Add your own custom styles below
.hero { background: linear-gradient(135deg, $primary, $secondary); }
```

Compile the Sass

```
# Install Sass
npm install -g sass

# Compile to CSS
sass custom.scss custom.css

# Watch for changes (auto-recompile)
sass --watch custom.scss:custom.css
```

5.4 Customizing Individual Components

Overriding Component Styles

```
/* custom.css — add after Bootstrap CSS */

/* Override button styling */
.btn-primary {
  background: linear-gradient(135deg, #667eea, #764ba2);
  border: none;
  border-radius: 50px;   /* Pill shaped */
  font-weight: 600;
  letter-spacing: 0.5px;
  box-shadow: 0 4px 15px rgba(102, 126, 234, 0.4);
}

/* Override card styling */
.card {
  border: none;
```

```
  border-radius: 1.25rem;
  box-shadow: 0 10px 30px rgba(0,0,0,0.08);
  transition: transform 0.3s ease;
}
.card:hover { transform: translateY(-5px); }

/* Override navbar */
.navbar {
  backdrop-filter: blur(10px);
  background: rgba(255,255,255,0.9) !important;
  box-shadow: 0 2px 20px rgba(0,0,0,0.1);
}
```

5.5 Bootstrap Utility API

Bootstrap 5 includes a powerful Utility API that allows you to generate custom utility classes using Sass maps — without writing repetitive CSS.

```
// Add custom utilities via the Bootstrap Utility API
@import 'bootstrap/scss/functions';
@import 'bootstrap/scss/variables';
@import 'bootstrap/scss/utilities';

// Add a custom 'blur' utility
$utilities: map-merge(
  $utilities,
  ('blur': (
    property: filter,
    class: blur,
    values: (sm: blur(4px), md: blur(8px), lg: blur(16px))
  ))
);

@import 'bootstrap/scss/utilities/api';

/* This generates: .blur-sm, .blur-md, .blur-lg classes */
```

Bootstrap's Utility API is extremely powerful for design systems. You can generate consistent spacing, color, and typography utilities that match your brand guidelines — all from one Sass configuration.

5.6 Bootstrap Themes & Resources

- **Bootswatch (free themes):** bootswatch.com — Drop-in Bootstrap themes (Cerulean, Darkly, Flatly, etc.)
- **Bootstrap Icons:** icons.getbootstrap.com — 1,800+ SVG icons
- **Bootstrap Studio:** Drag-and-drop visual Bootstrap builder
- **Official Docs:** getbootstrap.com/docs/5.3
- **Start Bootstrap:** startbootstrap.com — Free Bootstrap themes and templates

QUICK REFERENCE SUMMARY

RWD Concepts	Bootstrap Setup
• Fluid grids (% widths) • Flexible images • CSS media queries • Viewport meta tag • Mobile-first approach	• CDN link in <head> • JS bundle before </body> • npm install bootstrap • Import in SCSS/JS • Viewport meta required
Grid System	**Customization**
• container > row > col • 12-column system • 6 breakpoints (xs–xxl) • Gutter control (g-*) • Offset, order, nesting	• CSS :root variables • Sass variable override • Component class override • Utility API (Sass maps) • Bootswatch themes